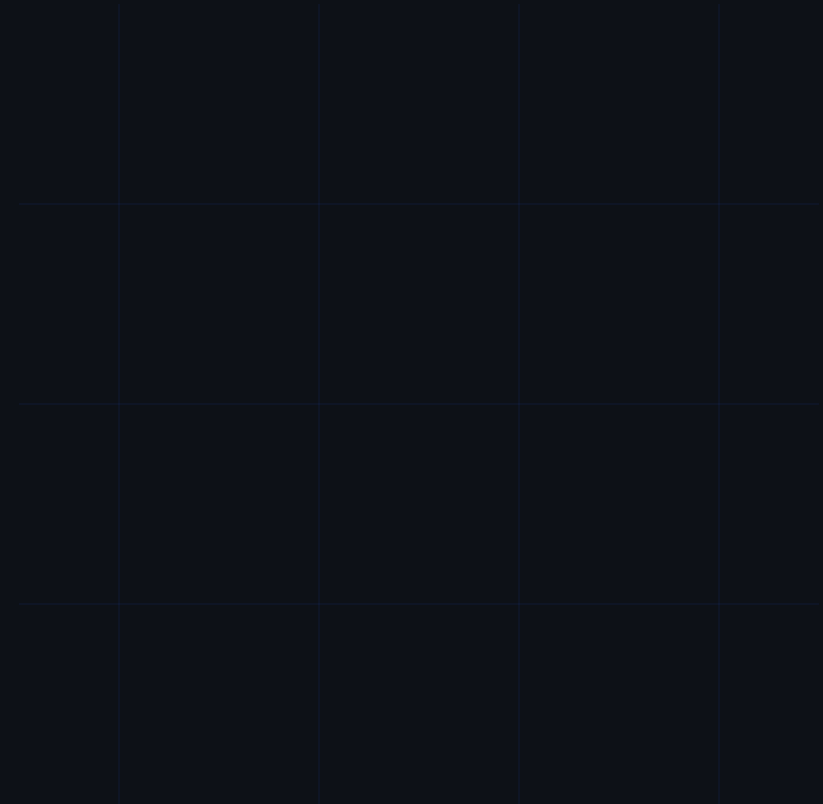


Global CO₂ Emissions Trends, Regional Shifts, and Per Capita Inequalities

A Data-Driven Briefing for Policy Analysts | 1750–2024

GLOBAL CARBON BUDGET 2025 · NASA



CO₂ concentration has risen 50%+ above pre-industrial levels — a change that took thousands of years now occurred in two centuries

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million (ppm)

NOAA Mauna Loa Observatory · February 2026

+53%

vs pre-industrial baseline (~280 ppm)

Context



Current level is at least 1.5× greater than the natural increase at the end of the last ice age 20,000 years ago.



That ancient change took thousands of years. The current change: roughly two centuries.



Driven primarily by burning coal, oil, and natural gas. Confirmed by ground-based and satellite measurements.

KEY INSIGHT

The speed of change is unprecedented. What nature accomplished over millennia, human activity has replicated in ~200 years. Ground-based and satellite observations confirm that fossil fuel combustion is the primary driver.

Global fossil fuel CO₂ emissions grew six-fold since 1950 — from 6 Bt to 35+ Bt — and have not yet peaked

Global Fossil Fuel CO₂ Emissions: 6 Billion to 35+ Billion Tonnes

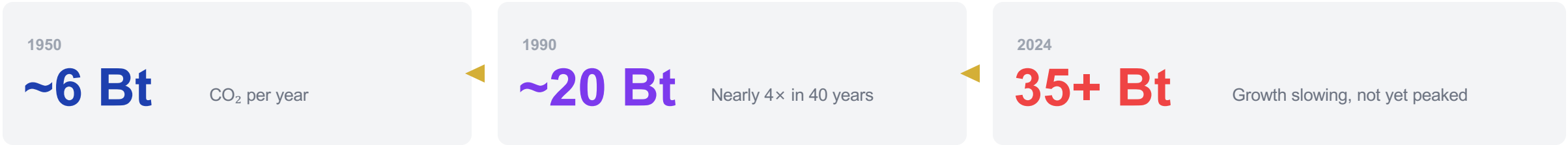
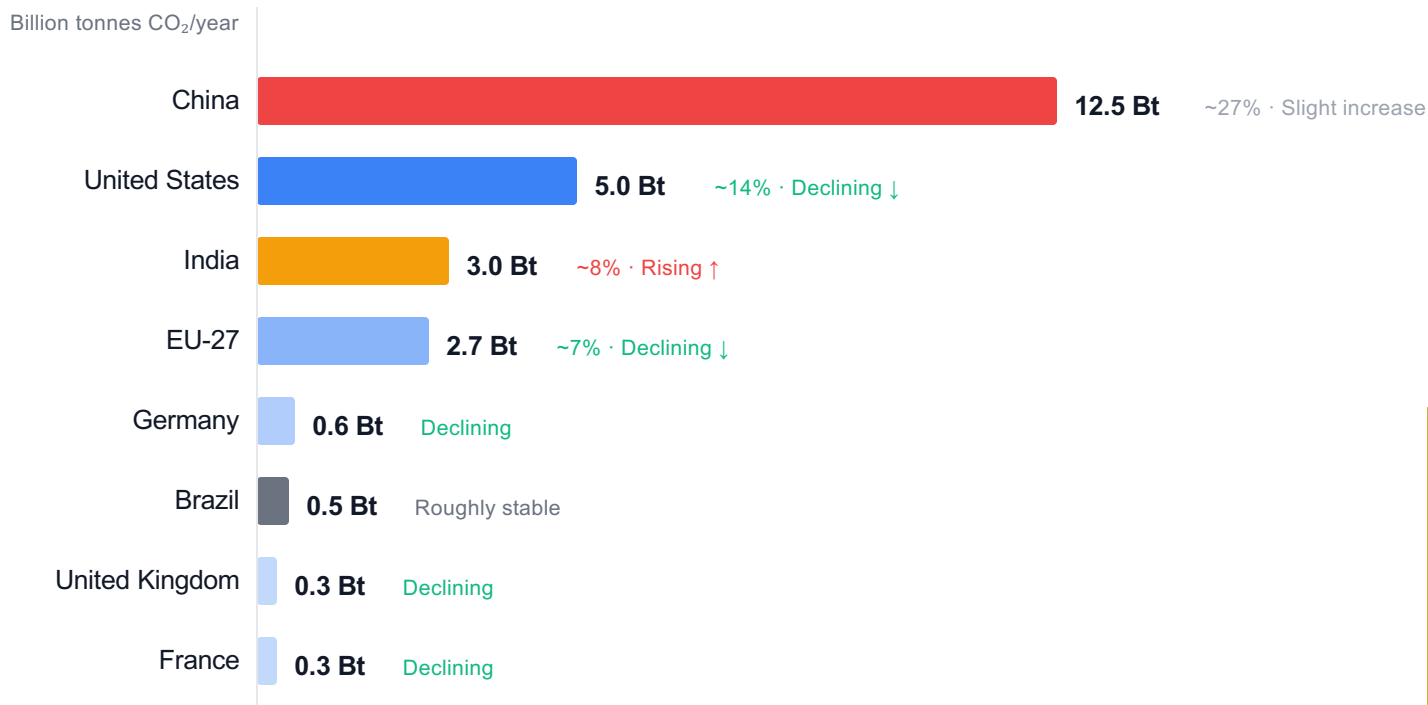


Figure: Annual CO₂ emissions by country, 1750–2024 (Our World in Data)

China now emits over 12 Bt annually — more than double the US (~5 Bt); India (~3 Bt) is third and rising fast

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

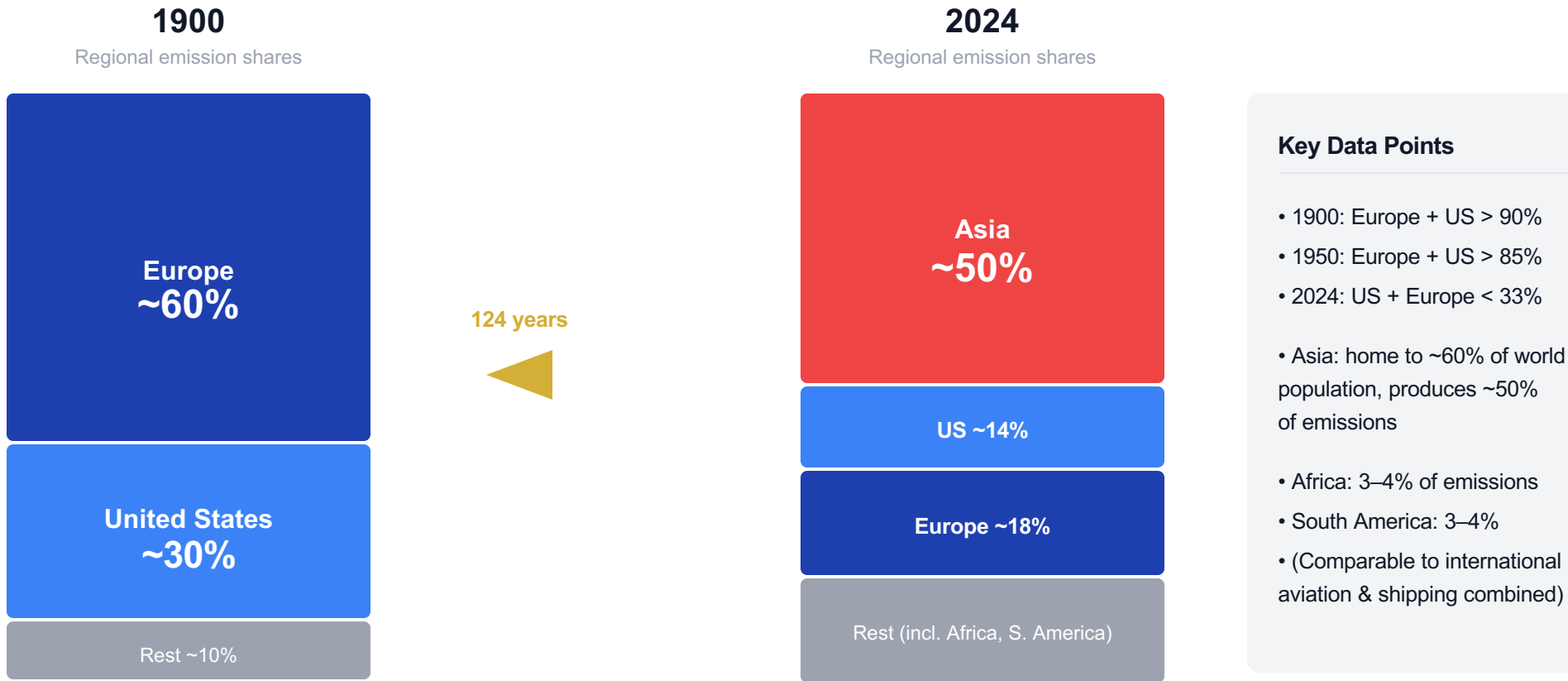


KEY CONTEXT

- China's emissions surged post-2000 with rapid industrialization
- US peaked near 6 Bt around 2005, now ~5 Bt
- India's trajectory is steepest among the top 3
- Europe's share has fallen substantially

The geographic center of emissions shifted dramatically — from >90% in Europe+US in 1900 to Asia producing ~50% today

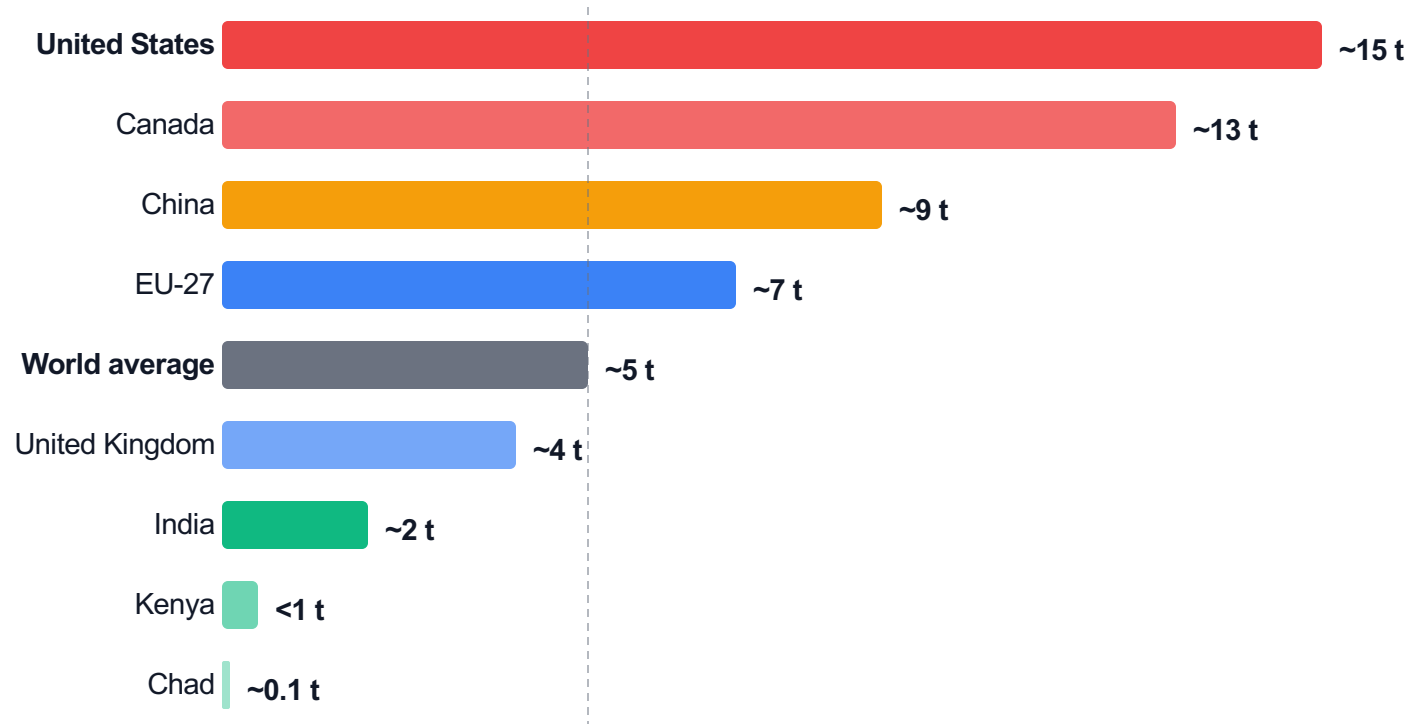
Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



A 150× disparity — the average American emits in under two days what a person in Chad emits in a year

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Tonnes CO₂ per person per year



150×

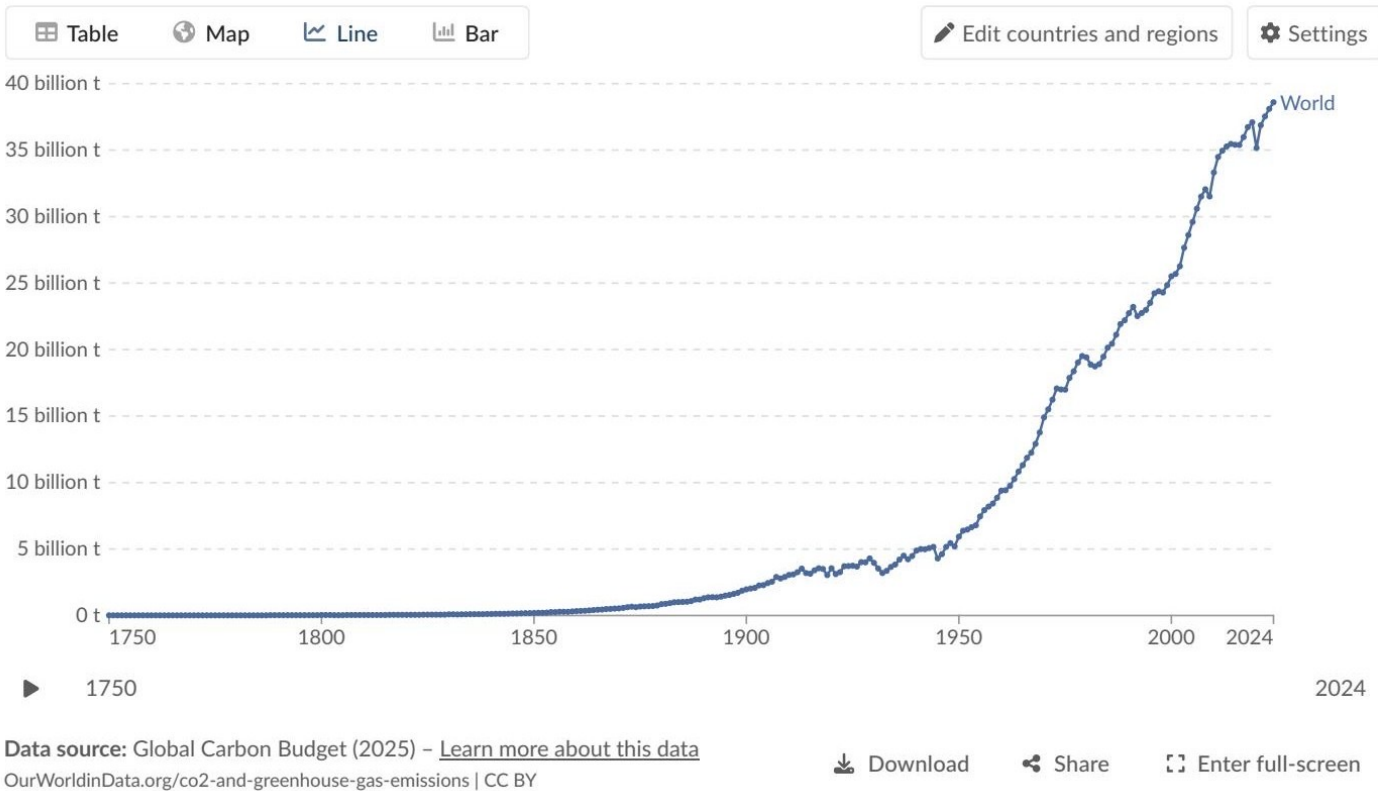
The average American emits in <2 days what a person in Chad emits in an entire year.

See figure_1.jpg for per capita trends over time

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.



Declining (Blue regions)

- United States: –1% to –3%
- Most of Europe: –1% to –5%
- Continued structural shift away from coal and heavy industry

Increasing (Orange/Red regions)

- Russia: +2% to +5%
- Southeast Asia: +3% to +8%
- Several African nations: +2% to +10%
- Some South American countries: >10%

IMPLICATION

Decarbonization progress in developed nations is being offset by growth in emerging economies.

Figure: Annual % change in CO₂ emissions, 2024

Countries with similar prosperity can have very different per capita emissions — energy mix and policy choices matter

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

All four countries below have comparable living standards and GDP per capita — yet per capita CO₂ differs by nearly 2×.

Country	Per Capita CO ₂	Primary Energy Sources	Key Factor	Trend
LOWER PER CAPITA EMITTERS				
France Western Europe	~4 t Below world avg (5 t)	High nuclear share (~70% of electricity) + growing renewables	Nuclear baseload	
United Kingdom Northern Europe	~4 t Below world avg (5 t)	Rapid coal phase-out, offshore wind growth, nuclear contribution	Coal phase-out + wind	
HIGHER PER CAPITA EMITTERS				
Germany Western Europe	~7+ t Above world avg	Heavy coal/lignite legacy, post-Fukushima nuclear phase-out, strong industry	Coal dependence	
Netherlands Western Europe	~8+ t Well above world avg	Natural gas dependent, large petrochemical & refining sector	Gas + industry	

 Similar prosperity, different emissions: energy mix and policy choices — not just wealth — determine per capita outcomes.

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting



Global emissions exceed 35 Bt/year and have not yet peaked

Despite slowing growth rates, the world continues to add more CO₂ to the atmosphere every year.



China alone accounts for >25% of the global total (~12.5 Bt)

More than double the United States (~5 Bt). India (~3 Bt) is rising fast as the third-largest emitter.



Geographic center shifted: Asia ~50%, US + Europe <33% (was >90% in 1900)

The map of responsibility has transformed dramatically in just over a century.



Per capita inequalities remain extreme — 150× gap

US/Australia at ~15 t/person vs. Chad at ~0.1 t/person. The average American emits in 2 days what Chad emits in a year.



Energy mix and policy choices — not just prosperity — determine outcomes

France/UK (~4 t) vs. Germany/Netherlands (~7–8 t): similar wealth, very different emissions due to energy policy.

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Source: Global Carbon Budget 2025, NASA, Our World in Data | April 2026